MAS

$\omega_0 = \sqrt{\frac{k}{m}}$

$x(t) = A\cos(\omega t + \varphi_0)$

$v(t) = -A\omega_0 \operatorname{sen}(\omega t + \varphi_0)$

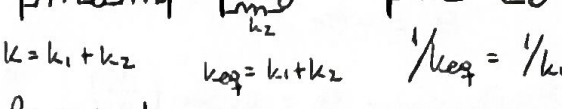
$a(t) = -A\omega_0^2 \cos(\omega t + \varphi_0)$

$A = \sqrt{x_0^2 + \left(\frac{v_0}{\omega_0}\right)^2}$

$\varphi_0 = \operatorname{arctg}\left(\frac{-v_0}{x_0\omega_0}\right)$ (chequear signo)

$E = \frac{1}{2}kA^2$ $\quad K(t) = \frac{1}{2}m v(t)^2$ $\quad U(t) = \frac{1}{2}kA^2\cos^2(\omega_0 t)$

Casos particulares



$k = k_1 + k_2$ $\quad k_{eq} = k_1 + k_2$ $\quad 1/k_{eq} = 1/k_1 + 1/k_2$

⊛ Si tengo un resorte de constante k y lo corto a la mitad ⇒ tengo 2 resortes de constante $k' = 2k$

Péndulo θ pequeño → $\theta \approx \operatorname{sen}\theta \approx \operatorname{tg}\theta$

$\theta(t) = A\cos(\omega_0 t + \varphi_0)$ $\quad \omega_0 = \sqrt{\frac{g}{L}}$ $\quad A = \sqrt{\theta_0^2 + \left(\frac{v_0}{L\omega_0}\right)^2}$ $\quad \varphi_0 = \operatorname{arctg}\left(\frac{-v_0}{\theta_0 L\omega_0}\right)$ (chequear resultado)

Amortiguado

$\gamma = \frac{b}{2m}$ $\quad \omega_0 = \sqrt{\frac{k}{m}}$, $\quad A = \sqrt{x_0^2 + \left(\frac{v_0 + \gamma x_0}{\omega'}\right)^2}$ $\quad \varphi_0 = \operatorname{arctg}\left(-\frac{v_0 + \gamma x_0}{x_0\omega'}\right)$

$F = -bv$

$\omega' = \frac{2\pi}{T'}$

$A(t) = A\cdot e^{-\gamma t}$

$E(t) \approx \frac{1}{2}mv^2 + \frac{1}{2}k\Delta x^2 \approx \underbrace{\frac{1}{2}kA^2}_{E_0} e^{-2\gamma t}$, $\quad \tau = \frac{1}{2\gamma}$

$\gamma > \omega_0$ → **sobreamortiguado**

$x(t) = e^{-\gamma t}\left(C_1 e^{\sqrt{\gamma^2 - \omega_0^2}\, t} + C_2 e^{-\sqrt{\gamma^2 - \omega_0^2}\, t}\right)$ ⊛ $v(t)$ atrás

$\gamma = \omega_0$ → **críticamente amortiguado**

$x(t) = (C_1 + C_2 t)e^{-\gamma t}$

$\gamma < \omega_0$ → **subamortiguado** $\quad \omega' = \sqrt{\omega_0^2 - \gamma^2}$

$x(t) = A e^{-\gamma t}\cos(\omega' t + \varphi_0)$

$v(t) = -Ae^{-\gamma t}\left[\gamma\cos(\omega' t + \varphi_0) + \omega'\operatorname{sen}(\omega' t + \varphi_0)\right]$

$Q = \omega_0 \tau = \frac{\omega_0}{2\gamma} = \begin{cases} <1/2 \to \text{sobre} \\ = 1/2 \to \text{crit} \\ > 1/2 \to \text{sub} \end{cases}$

$\omega' = \sqrt{1 - \frac{1}{4Q^2}}\cdot\omega_0$

$\gamma T' = \frac{\pi}{\sqrt{Q^2 - 1/4}}$

$\frac{\Delta E(t+T')}{E(t)} = \frac{2\pi}{Q}$ si Q grande

$\frac{\Delta E}{E} = \frac{E(t+T') - E(t)}{E(t)} = e^{-2\gamma T'} - 1$

$|E_{sf}| = 1 - e^{-2\gamma T'}$ fracción de E que se pierde en cada ciclo

período de $\operatorname{tg} = \pi$

$\operatorname{arctg}(\ldots) = \quad \operatorname{tg}(x) = n \Rightarrow x = \operatorname{arctg}(n) + k\pi$

Forzado

$F(t) = F_0\cos(\omega t)$ $\quad A_{est} = \frac{F_0/m}{\sqrt{(\omega_0^2 - \omega^2)^2 + (2\gamma\omega)^2}}$ $\quad \operatorname{tg}\delta = \frac{2\gamma\omega}{\omega_0^2 - \omega^2}$ $\quad \operatorname{sen}\delta = \frac{2\gamma\omega}{\sqrt{(\omega_0^2 - \omega^2)^2 + (2\gamma\omega)^2}}$

$x_p(t) = A_{est}\cdot\cos(\omega t - \delta) + \varphi_0$

$x_h(t)$ igual que amortiguado

$\omega_{rA} = \sqrt{\omega_0^2 - 2\gamma^2}$ $\quad A_{max} = A(\omega_{rA}) = \frac{F_0}{2\gamma m\sqrt{\omega_0^2 - \gamma^2}} = \frac{F_0}{b\omega'}$ $\quad \omega_{rP} = \omega_0$

$\Delta\omega = \frac{\omega_0}{Q} = 2\gamma$ → Cuando $A = \frac{A_{max}}{2}$, $\langle P\rangle = \frac{\langle P\rangle_{max}}{2}$

$\langle P\rangle = \frac{F_0^2}{4m\gamma}\cdot\frac{(2\omega\gamma)^2}{(\omega_0^2 - \omega^2)^2 + (2\gamma\omega)^2}$, $\quad \langle P\rangle_{max} = \langle P\rangle_{(\omega_{rP})} = \frac{F_0^2}{4m\gamma}$

$\langle \operatorname{sen}^2 x\rangle = \langle\cos^2 x\rangle = \frac{1}{2}$

Ondas propagantes

$\omega = kv$ $\quad v = \frac{\lambda}{T} = \lambda f$ $\quad \lambda = \frac{2\pi}{k}$ $\quad \mu = \frac{\Delta m}{\Delta x}$ $\quad v = \sqrt{\frac{F}{\mu}}$ $\quad \langle P_{(x,t)}\rangle = \frac{1}{2}\mu\omega^2 A^2 v$

$y(x,t) = A\cos(kx \pm \omega t + \varphi_0)$

$v_y(x,t) = \mp A\omega\operatorname{sen}(kx \pm \omega t + \varphi_0)$

$a_y(x,t) = \pm A\omega^2\cos(kx \pm \omega t + \varphi_0)$

$u_k = \frac{1}{2}\mu\left(\frac{\partial y}{\partial t}\right)^2 = \frac{1}{2}\mu\omega^2 A^2\operatorname{sen}^2(kx \pm \omega t + \varphi_0)$, $\quad \Delta k = u_k\Delta x$

$u_U = \frac{1}{2}F\left(\frac{\partial y}{\partial x}\right)^2 = \frac{1}{2}Fk^2A^2\operatorname{sen}^2(kx \pm \omega t + \varphi_0)$, $\quad \Delta U = u_U\Delta x$

$Fk^2 = \mu\omega^2 \Rightarrow u_U = u_k \to u_E = 2u_k = 2u_U$, $\quad P = u_E v$

$\langle u_U\rangle = \langle u_k\rangle = \frac{1}{4}\mu A^2\omega^2$, $\quad \langle u_E\rangle = 2\langle u_U\rangle$, $\quad \langle \operatorname{sen}^2 x\rangle = \frac{1}{2}$, $\quad E = \langle P\rangle t$ $\quad I = \frac{\langle P\rangle}{S}$ (Intensidad = Potencia / Superficie)

Ondas estacionarias

$A = 2A'$ (2 veces la de las propa) , misma v que propagantes

$y(x,t) = \underbrace{A\cos(kx + \varphi_x)}_{A(x)}\cdot\underbrace{\cos(\omega t + \varphi_t)}_{f(t)}$

$v_y(x,t) = -A\omega\cos(kx + \varphi_x)\operatorname{sen}(\omega t + \varphi_t)$

$a_y(x,t) = -A\omega^2\cos(kx + \varphi_x)\cos(\omega t + \varphi_t)$

f — f $\quad \lambda_n = \frac{2L}{n}$, $\quad f_n = v\frac{n}{2L}$, $\quad n \in \mathbb{N}$ (A)

f — l $\quad \lambda_n = \frac{4L}{n}$, $\quad f_n = v\frac{n}{4L}$, $\quad n$ impar (B)

l — l $\quad \lambda_n = \frac{2L}{n}$, $\quad f_n = v\frac{n}{2L}$, $\quad n \in \mathbb{N}$ (C)

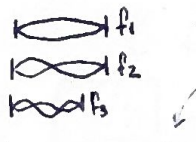
libre → antinodo → máxima amplitud

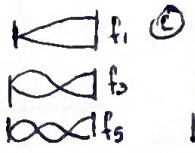
fijo → nodo → $y = 0\ \forall t$

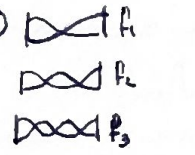
cuerda horizontal → $k = k_{max}$ $\quad U = 0$

máx elongación → $k = 0$, $\quad U = U_{max}$

$\int \operatorname{sen}^2(ax)\,dx = \frac{x}{2} - \frac{\operatorname{sen}(2ax)}{4a}$ | $\cos^2$ igual pero con +

(A) 

(B) 

(C) 

$\cos(kx + \varphi_x) = \operatorname{sen}(kx)$ $\quad \cos(kx + \varphi_x) \to \cos(kx)$

si $y = A\operatorname{sen}(kx)\cos(\omega t + \varphi_t)$ (X)

$u_k = \frac{1}{2}\mu\left(\frac{\partial y}{\partial t}\right)^2 \overset{x}{=} \frac{1}{2}\mu\omega^2 A^2\operatorname{sen}^2(kx)\operatorname{sen}^2(\omega t + \varphi_t)$ $\quad (Fk^2 = \mu\omega^2)$

$u_U = \frac{1}{2}F\left(\frac{\partial y}{\partial x}\right)^2 \overset{x}{=} \frac{1}{2}Fk^2A^2\cos^2(kx)\cos^2(\omega t + \varphi_t)$

$u_E = u_k + u_U \overset{x}{=} \frac{1}{2}\mu\omega^2 A^2\left[\operatorname{sen}^2(kx)\operatorname{sen}^2(\omega t + \varphi_t) + \cos^2(kx)\cos^2(\omega t + \varphi_t)\right]$, $\quad E = \int_{x_i}^{x_f} u_E(x,t)\,dx$

si $\begin{cases} \Delta x = x_f - x_i = n\frac{\lambda}{2}, \ n \in \mathbb{Z} \ \text{o} \\ \Delta x = n\frac{\lambda}{4} \ \text{con} \ x_i \ \text{nodo/antinodo} \end{cases}$ $\Rightarrow E = \frac{1}{2}\mu\omega^2 A^2\frac{\Delta x}{2}$

(osea integrar entre nodos/antinodos)

fuera si se cumple

$\int_{x_i}^{x_f}\cos^2(kx)\,dx = \int_{x_i}^{x_f}\operatorname{sen}^2(kx)\,dx = \frac{\Delta x}{2}$

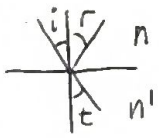
$\langle P\rangle = 0$ $\quad P_{(x,t)} = \frac{1}{4}\mu\omega^2 A^2 v\operatorname{sen}(2kx)\operatorname{sen}(2\omega t + 2\varphi_t)$

$\langle u_k\rangle = \langle u_U\rangle = \frac{1}{4}\mu\omega^2 A^2 = \frac{1}{4}Fk^2A^2$

$\langle u_E\rangle = \frac{1}{2}\mu\omega^2 A^2$

Óptica geométrica

$C = 3\cdot10^8 \frac{m}{s}$ $\quad n = \frac{c}{v}$ ← velocidad de la luz en el medio $\quad n_{vacío} = 1 \approx n_{aire}$



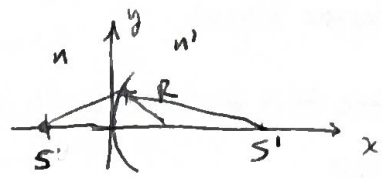
Snell: $n\,sen(i) = n'sen(t)$

$\theta_i = -\theta_r$

$n' > n \rightarrow \theta_t < \theta_i$ (se acerca a la normal)

$n' < n \rightarrow \theta_t > \theta_i$ (se aleja de la normal)

$\lambda = \frac{\lambda_0}{n}$ $\quad$ ángulo crítico $\theta_c = arc\,sen\left(\frac{n'}{n}\right)$



aprox parax: $sen\theta \approx \theta \approx tg\theta$

Dioptras

$$\frac{n'}{s'} - \frac{n}{s} = \frac{n'-n}{R} = \phi = \frac{n'}{f'} = -\frac{n}{f}$$

$$A = \frac{y'}{y} = \frac{ns'}{n's}$$

Dioptra plana $R \to \infty$, $\frac{n'}{s'} = \frac{n}{s}$

convergente	divergente
$f' > 0$	$f' < 0$
$f < 0$	$f > 0$

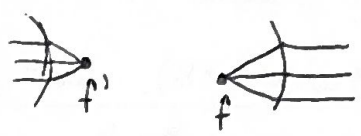
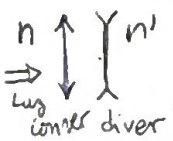


imagen real → convergen en un punto
imagen virtual → no

Lentes

$$\frac{n'}{s'} - \frac{n}{s} = \frac{n_L - n}{R_1} + \frac{n' - n_L}{R_2} = \phi = \frac{n'}{f'} = -\frac{n}{f}$$

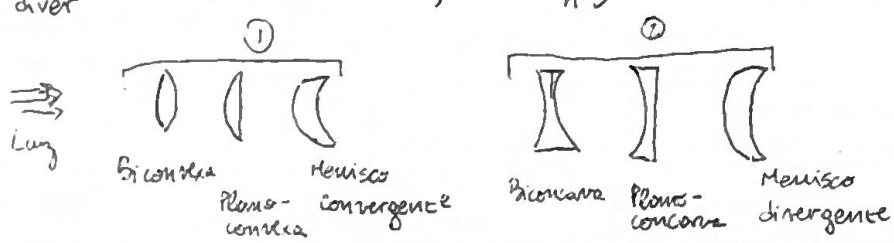
si $n = n' \rightarrow \frac{1}{s'} - \frac{1}{s} = \left(\frac{n_L}{n} - 1\right)\left(\frac{1}{R_1} - \frac{1}{R_2}\right) = \phi = \frac{1}{f'} = \frac{-1}{f}$



$\cdot A = A_1 A_2 \dots A_n$, $\quad A = \frac{ns'}{n's}$

Varias lentes $\quad s_2 = s_1' - D$

$A = \frac{s'}{s}$



Ø $n_L > n, n'$
① convergente
② divergente

$n_L < n, n'$
① divergente
② convergente

espejo plano → $R \to \infty$

$\frac{1}{s'} + \frac{1}{s} = \frac{2}{R} = \frac{1}{f} = \frac{1}{f'}$

$A = -\frac{s'}{s}$

cambia dir de los rayos → cambiamos ⊕

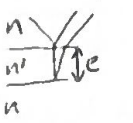
Interferencia y difracción

$I = I_1 + I_2 + 2\sqrt{I_1 I_2}\,\langle \cos\alpha \cos\Delta\phi \rangle$. $\quad I_0 = \frac{nc\varepsilon_0}{2} A^2$ $\quad \varepsilon_0 = 8{,}845\cdot10^{-2} \frac{ws}{mv^2}$ $\quad [I] = \frac{W}{m^2}$

$\Delta\phi = \frac{2\pi}{\lambda_0}\Delta CO + \Delta\varphi \rightarrow \Delta\phi = 2k\pi$ constr, $\Delta\phi = (2k+1)\pi$ dest (Un haz se invierte cuando se refleja con medio de n mayor $\pm\pi$)

División de amplitud

Láminas delgadas



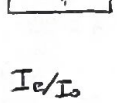
$\Delta\phi = \frac{2\pi}{\lambda_0} n' 2e + \Delta\varphi$

Cuña $e = x\,tg\alpha \approx \alpha x$



$\delta x = \frac{\lambda_0}{2n\alpha}$ $\quad \Delta\phi = \frac{2\pi}{\lambda_0} 2n\alpha x + \Delta\varphi$

Anillos de Newton



$\Delta\phi = \frac{2\pi n r^2}{\lambda_0 R} + \Delta\varphi$

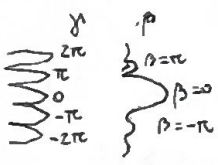


$\Delta\phi = \frac{4\pi ne}{\lambda_0} - \frac{2\pi n r^2}{\lambda_0 R} + \Delta\varphi$

División de frente de onda (Doble rendija)

$\gamma = \frac{\Delta\phi}{2} = \frac{\pi}{\lambda_0}\Delta CO$ $\quad \beta = \frac{ka\,sen\theta}{2} = \frac{\pi a}{\lambda_0} n\,sen\theta$

$I = 4I_0 \cos^2\gamma \left(\frac{sen\beta}{\beta}\right)^2$



β	I_θ/I_0	
0	1	max
π	0	min
1,4303π	0,047	max
2π	0	min
2,4590π	0,016	max

$\gamma = m\pi \rightarrow$ max
$\gamma = (m+\frac{1}{2})\pi \rightarrow$ min
$\beta = p\pi \rightarrow$ min

$m = p\frac{d}{a}$ → Suprime max m con mínimo p

si $m \in \mathbb{Z}$ $p \in \mathbb{Z}$

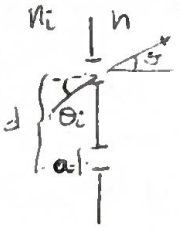
Desplaza todo → $\Delta y = \frac{\lambda_0 S}{nd}$

Desplaza interferencia → lámina delgada → Si lámina abajo $(n'-n)$



$\gamma = \frac{\pi}{\lambda_0}[nd\,sen\theta + (n-n')e]$

$\beta = \frac{\pi}{\lambda_0} an\,sen\theta$



incidencia paralela

$\gamma = \frac{\pi}{\lambda_0} d\,n\,sen\theta$

$\beta = \frac{\pi}{\lambda_0} a\,n\,sen\theta$

incidencia oblicua

$\gamma = \frac{\pi}{\lambda_0} d(n\,sen\theta - n_i\,sen\theta_i)$

$\beta = \frac{\pi}{\lambda_0} a(n\,sen\theta - n_i\,sen\theta_i)$

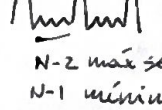
Babinet: abertura de ancho a difracta lo mismo que un obstáculo de ancho a

Redes de difracción

N rendijas iguales y equiespaciadas (ancho a, distancia d) $\quad K = \frac{1}{d}$

$$I = I_0 \left(\frac{sen(N\gamma)}{sen\,\gamma}\right)^2 \left(\frac{sen\beta}{\beta}\right)^2$$

$\boxed{sen\theta = mk\lambda}$ $\quad \lambda = \frac{\lambda_0}{n}$

interf: 

N-2 máx secundarios
N-1 mínimos

máximos ppales → $\frac{I_0}{N}$ → $\boxed{N^2 I_0} = I_{max}$

cuando → máximos en $m\pi$

los puedes suprimir los mínimos de difracción $m = p\frac{d}{a}$

Para que λ_1, λ_2 estén resueltas $\quad \Delta\lambda \geq \Delta\lambda_{min} = \frac{\lambda_1}{mN}$

Resolución $\quad \mathcal{R} = \frac{\lambda_1}{\Delta\lambda_{min}} = mN$, $\quad \lambda_1 < \lambda_2$

ancho del haz: $W = \frac{N}{K} = Nd$

Sobre amortiguado

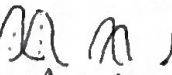
$r(t) = C_1 e^{t(\Gamma - \gamma)}(\Gamma - \gamma) + C_2 e^{t(-\Gamma - \gamma)}(-\Gamma - \gamma)$

$\Delta CO = n_1 z_1 - n_2 z_2$ en los de doble rendija es rayo de abajo − rayo de arriba

Rayleigh: resueltas si máximo coincide con mínimo

$\theta_{min} = \frac{\lambda}{a}$

$\theta_{min} = 1{,}22\frac{\lambda}{a}$ (circular)



resueltas no resu

— 0

Termodinámica

Unidades

$K = °C + 273{,}15\,K$

$1\ cal = 4{,}186\ J$

$1\ atm = 101\,325\ Pa$

$Pa = N/m^2$

$1\ l \cdot atm = 101{,}325\ J$

$1\ mL = 1\ cm^3$

$1\ L = 0{,}001\ m^3$

Ley 0

Dos sistemas en equilibrio térmico sii $T_1 = T_2$

Escala térmica → $T = \frac{L - L_0}{L_{100} - L_0}\,100°C$

$T = \frac{P - P_0}{P_{100} - P_0} \cdot 100°C$

$Q = mc\Delta T$

si hay cambio de fase → $Q = mL$

Sustancia	cal/g°C	J/kg°C
Agua	1	4200
hielo	0,5	2100
Vapor	0,48	2016

Expansión térmica



(1D) $\Delta L = \alpha L_0 \Delta T$

(2D) $\Delta A = 2\alpha A_0 \Delta T$

(3D) $\Delta V = 3\alpha V_0 \Delta T = \beta V_0 \Delta T$

α: coef de expansión lineal, $[\alpha] = K^{-1}$

Esfuerzo térmico

$\frac{F}{A} = Y\alpha\Delta T$

Y: medida de presión propia del material

Conducción térmica

$\phi = \frac{Q}{\Delta t} = \frac{\Delta T}{L} A k$

ϕ flujo de calor $[\phi] = \frac{J}{s} = W$

k: conductividad $[k] = \frac{W}{mK}$

$L_f = 80\ \frac{cal}{g} = 334\ kJ/kg$

$L_v = 540\ cal/g = 2264\ kJ/kg$

1era Ley

⊕ $Q > 0$ → sis recibe calor ⊕ $W > 0$ → sis hace trabajo

⊖ $Q < 0$ → sis cede calor ⊖ $W < 0$ → sis recibe trabajo

$$\Delta U = Q - W$$

Gases ideales

$PV = nRT$ $\quad R = 0{,}082\ \frac{l \cdot atm}{K\,mol} = 8{,}31\ \frac{J}{K \cdot mol}$

monoatómico → $U = \frac{3}{2}nRT$ $C_v = \frac{3}{2}R$ $C_p = C_v + R = \frac{5}{2}R$

diatómico → $U = \frac{5}{2}nRT$ $C_v = \frac{5}{2}R$ $C_p = C_v + R = \frac{7}{2}R$

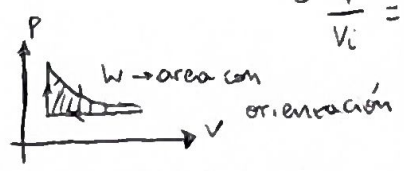
Trabajo para cambiar el volumen

$dW = P_{ext}\,dV \rightarrow W = \int_{V_i}^{V_f} P_{ext}\,dV$

equilibrio y procesos reversibles → $P_{ext} = P$

~~iso~~ P(V) depende del proceso.

- isocórico ($V = cte$)
  $dW = PdV = 0$, $\Delta U = Q = nC_v\Delta T$
- isobárico ($P = cte$)
  $W = P\Delta V$, $Q = \Delta U = nC_p\Delta T$
- isotérmico ($T = cte$)
  $dV = nC_v dT = 0$, $Q = W = nRT\ln\left(\frac{V_f}{V_i}\right)$
  ⊛ $\frac{V_f}{V_i} = \frac{P_i}{P_f}$
- adiabático ($dQ = 0$)
  $\gamma = \frac{C_p}{C_v}$, $TV^{\gamma-1} = cte$, $PV^{\gamma} = cte$
  $\Delta U = nC_v\Delta T = -W$, $W = \frac{1}{\gamma - 1}(P_iV_i - P_fV_f)$
- cíclico ($\Delta U = 0$)
  $Q = W = \Sigma Q_i = \Sigma W_i$



2da Ley

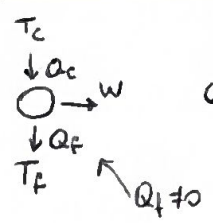
Cíclicos ⇒ $\Delta U = 0$

⇒ $W = Q = Q_f + Q_c$

Maquina térmica

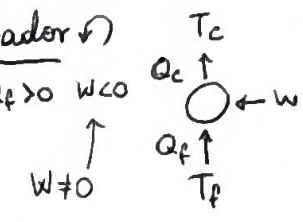
$Q_c > 0$ $Q_f < 0$ $W > 0$

$e = \frac{W}{Q_c} = 1 - \frac{|Q_f|}{|Q_c|}$



Refrigerador

$Q_c < 0$ $Q_f > 0$ $W < 0$



$k = \frac{|Q_f|}{|W|} = \frac{|Q_f|}{|Q_c| - |Q_f|}$

Carnot → cotas de eficiencia

$e_{carnot} = 1 - \frac{T_f}{T_c}$, $\frac{T_f}{T_c} = \frac{|Q_f|}{|Q_c|}$

$k_{carnot} = \frac{T_f}{T_c - T_f}$

- isotermico → $\Delta S = nR\ln\left(\frac{V_f}{V_i}\right)$
- isocórico → $\Delta S = nC_v\ln\left(\frac{T_f}{T_c}\right)$
- isobárico → $\Delta S = nC_p\ln\left(\frac{T_f}{T_i}\right)$

Entropía

$dS = \frac{dQ_{rev}}{T} \Rightarrow \Delta S = \int_i^f \frac{dQ_{rev}}{T}$

$\Delta S_U = \Delta S_{sis} + \Delta S_{ent} \geq 0$

$[\Delta S] = J/K$

- adiabáticos → $\Delta S = 0$
- cambio de estados ($T = cte$) → $\Delta S = \pm\frac{mL}{T}$

Maq T.
$\Delta S_{ent} = -\frac{|Q_c|}{T_c} + \frac{|Q_f|}{T_f}$

Refri
$\Delta S_{ent} = \frac{|Q_c|}{T_c} - \frac{|Q_f|}{T_f}$

ΔS_{sis} cíclico es 0